Bradly Alicea: "An Overview of Open Source and Open Science"

51:44

all right the next recorded talk the
penultimate recorded talk is by Bradley
Alysia
earlier Bradley gave a talk on
purposefully non-standard
cybernetics and this talk is going to be
an overview of Open Source and open
science so that's really cool and thanks
for bringing this important
complimentary topic to the Symposium for
all all all who are working on this okay
it's 51 minutes
long hello my name is Dr Bradley Alisa
and I'm a member of the active inference
Institute scientific Advisory Board
today I'm going to give you an overview
of Open Source and open science I'm my
home institution is the orthogonal
research and education lab I'm also
affiliated with the openworm foundation
and the University of Illinois champagne
or champagne
I'm also with particular interest in the
orthogonal research and education lab is
our open source interest group and I'll
talk about that throughout this talk
this starts with a question what can we
do with open source and open
SS and I pair these together because
sometimes we're building software
sometimes we're doing scientific
discovery which kind of relies on the
open source software and so if we're
doing both we can actually build uh
systems that can sort of help us do

better software engineering and science
so they're kind of grouped together so
in terms of Open Source you know it it's
maybe a truism that open source is about
software and putting it out into the
world but that's not enough and you
really want to what you want to do
instead of just putting software into
the world is building a community around
technical standards and code and this is
true if you want to build a community of
users or an open source community that
helps build the software
because you need to have some
interaction with the software to make it
sustainable you also want to have good
technical standards and with academic
software as we see that's been lacking
this also allows us to open up the
organization's processes and products so
having these communities and having
these technical standards are not only
good from just just stamping of sharing
software but it also opens up our
processes and products and allows us to
serve our communities better our
academic community
ities open science is where we share
data share results share research
artifacts of the public so we do a lot
of sort of transparency related things
we give people our results and our data
and it's software and papers and
presentations and we're very open about
we can give them access to these things
so when I say open science I also kind
of mean Open Access and it's they're
kind of in the same bucket

open science in general allows us to
open up our organization's research
capabilities so that people can benefit
from our discoveries and we can benefit
from their
feedback so a lot of organizations have
propose different sort of road maps or
statements on open science and what that
involves so not every organization's
view of this is going to be the same but
I'm using the young European Arch
University's Network statement on open
science at the link below and to them
you know they have a very strong
emphasis on collaboration and citizen
science so this is where you have
collaboration around the world around
the scientific
community and also involving say citizen
science where people just want to
contribute their own um analyses
sometimes it's data sometimes it's
observations things like that there's
visibility and idea sharing so know
being able to get to the outputs and
read them and understand them is
important open education is important
idea sharing because you need to have
you know uh focus on
collaboration alternative models of
publishing you need to be able to
publish the work to get it out there
into people's
hands incentives assessments and
research Integrity so those are all kind
of part of this open science tree now
again every organization who Champions
open science will have a different view

of this we don't really have a
formalized view in the orthogonal lab
but it follows this kind of outline
close quite
closely and so if you want to know more
about the where the open or where the
orthogonal research lab is coming from
you can read this paper and I put this
out in 2020 he needs to be updated but
it's a good
overview and this is building and
distributed virtual Laboratory adjacent
to
Academia um and and so there are a
number of things in this paper it's just
basically it's it's very similar to the
model that the active inference lab uses
were much more focused on sort of
educational opportunities and
opportunities for people to sort of pick
up the load and uh
build uh research
artifacts and you know having students
come in and do short-term projects but
it's it's it's a very similar process so
one of the processes or one of our goals
is to integrate open access in our open
source practices for advancing
distributed research projects so we want
to be able to help people make these
things open help them build research
projects the other thing is that we want
to encourage practice on open working
through the creation of digital research
artifacts it's like papers or
presentations and when I say open
working I mean I borrow this term from
uh the Mozilla foundation and they're

approach to this whole Enterprise is
basically working open so it's working
in these collaborative distributed
groups and how to make the most of that
kind of
experience you know things like note
taking and meetings and managing sort of
the human capital so that's also part of
it open source and open Sciences
managing human capital managing these
work
groups so one of the big thing other big
things about open source and open
science this idea
practice and so you know it's it's very
practice driven as opposed to sort of
theoretical or philosophically driven
and so you know practice means that you
need to establish
practices procedures routines workflows
and Technical Solutions for working open
so this means you need to adopt open
practices it could be Community review
of code releasing code under a license
printing
manuscripts and generating open data
sets so these are all things that
involve you know planning but also
practice and being able to produce
something at the
end related to that are what we call
open workflows so when we want to
release an open data set we can't just
pop data out onto the internet we have
to put open data sets into a production
line model where we you know clean the
data
we analyze the data we produce metadata

and then we publish that formally in a
repository with metadata
attached we also want to have Community
standards for open participation so we
can't just expect people to pick things
up and run with them without giving them
some standards for working so you know
you know usually in academic work you're
trained a certain number of years it
becomes kind of intuitive to you what to
do but a lot of people especially
students
undergraduates they come in and they
don't know what to do they don't have
like those standards ingrained and so
you have to have Community standards but
also because sometimes the incentives
aren't there for people to follow long
lengthy procedures and so they
don't in any case this helps you
standardize this e these efforts then we
have open Technical Solutions so open
source and open science are built or
enabled by technical solu Solutions you
couldn't really have open source and
open science without having open
Technical Solutions like GitHub live
streaming tools and bibliographic tools
and of course those things don't make
something open source or open science by
themselves but they enable those
practices so this is a nice graph where
I kind of talk about vision of
sustainable open science and so I have a
couple of categories here to the left
and the right which are are sort of
suboptimal areas and then where we want
to go is we want to go to this Zone this

elusive Zone in the red box so the
elusive Zone the red box are the set of
tools that would allow us to do
sustainable open science and Open Access
and the stuff to the left and the right
are kind of like things that we do now
that are suboptimal so we start with the
suboptimal and we talk about black Open
Access which is kind of an fancy word
for piracy or informal sharing and so
each of these types of open aess are
colored black just means it's kind of
related to under the table or you know
piracy or something so one of the
examples are is syab where you know it's
I my position on piracy is it's
Justified given the incentive structure
in science but you know you have systems
where you can just get access to things
and then of course sometimes we share
PDFs by sending emails to our colleagues
and sending them the paper and that's
all black Open Access because it's very
informal and it's not really sustainable
because you're relying on people to have
you know a you know someone has to get
access to these and then share it with
someone
else um then you go over to the other
side you have two other areas of Open
Access you have gold Open Access and
diamond platinum Open Access so gold
open and access is where typically an
author will pay APC fee which is a
publication fee it's usually a fairly
decent sum of money to a publisher to
make a paper open access to all L of
tend journals will restrict access to

paying customers if you are member of an
academic Library you can get access but
if you're not then you don't have access
and that's one of the things about like
places like the active inference
Institute or the orthogonal lab is not
everyone has that academic Library
access and so it's really hard to get
things get access to papers and things
that you need so this becomes a very
important set of points
here gold Open Access then of course is
is beneficial to the Publishers but not
to the authors diamond platinum Open
Access is more beneficial to the authors
there's no APC fee but someone has to
take that fee and you know they have to
absorb that fee so that's not optimal
either so this elusive zone is a way to
sort of build on sort of the
lowcost but still like sufficient
resources we need to provide access to
our work we need to open it up give
everyone access to it and allow them to
interact with the work in a way we
couldn't even do with any of these other
types of Open Access so the first thing
we have is green Open Access which is
pre-printing so this is where it's like
not the gold standard but it's also you
know you can post a paper on a preprint
server there's a pre-print server that
it's funded through like a foundation
and you know that's that makes it open
Endeavor um they're you know formal sort
of you know there's a DOI which is a
formal link it's a permanent link so
your work is in a place that has you

know some level of reputation with with
a mark of approval on it you can also
take that preprint and go through what's
called open peer review where you open
up your paper to peer review not to like
three referees you've never met but to
the entire community and the thing about
thing about peer review and I won't get
into this uh any more than what I'll
talk about here but if we do peer review
with like three people on an editor you
know you pick the people who are maybe
the most available or maybe the most
visible people in the field or maybe
people with the most time and they have
certain opinions about what you should
do what you should add on to a paper to
it
publishable and sometimes it's not in
the best interest of the
science
furthermore given that they still don't
really have a very good handle on
detecting things like mistakes or even
fraud so you can have a paper that goes
through three reviewers and an editor
and still be retracted for mistakes or
fraud so that's not good either open
peer review allows us to have people
many eyes on the paper many comments you
can incorp corporate the comments by
generating a new version of the preprint
and you have this continual feedback of
your work and you can eventually maybe
publish it somewhere but you can also
update the pre-print
continually and have that process going
um so that's you know Green Open Access

and open peer review now a lot of papers
also have data and we can include open
data management in this elusive Zone
because we also want to have a strategy
for managing and sharing data and this
goes along with the peer review and the
access of the preprint that that's the
link to it and that the data can then be
reviewed and worked with independently
of the preprint but also you know it
it's transparent and people can actually
work with it far beyond the LIF span of
a single
paper then we have overlay journals so
we can have preprints we can have but
that's it's on a pre-print server we can
have mulp versions but sometimes we want
to build our own journals because we
want to narrow the focus of the
preprints funnel them down
into things of interest for certain
Fields so say for example we had an
active inference overl joural it would
take pre-prints in the active inference
area only of those pre-prints and
present them as Journal content so what
journals do besides conferring status on
people is they give you this curation of
the literature of of research so that's
what overlay journals can do but without
the overhead of a journal so you can
have like your preprints you put them on
this with the skin over the top that
gives us this emperator
of uh you know what a journal would give
you so that's the elusive zone of Open
Access um and I just wanted to point
that out because I think it's you know a

good way to think about sort of the
production of Science and the production
of academic work
I posted a blog post for Open Access
week
2024 and it's kind of a critique of Open
Access as it exists and this is why I
kind of went through this and basically
I argu in this post that open access is
essentially failed at least up to this
point that we don't have the right
incentive structures we don't have the
right intrinsic and extrinsic incentives
to do open science correctly Open Access
and so we need to change those and so
you can read that post on my blog
synthetic daisies so in that blog post I
talk about the intrinsic and extrinsic
motivators for doing Open Access
sustainable so this you think of this
more generally as a reinforcement
learning of working open or
reinforcement learning of so when I talk
about intrinsic motivators I talk about
the internal drive to do something so
you know the idea is that you would do
things that are meaningful to yourself
that's an intrinsic motivator
or what motivates me in terms of a
scientific question what motivates me in
terms of how I can contribute to the
field and then how can I benefit from or
contribute to a team these are all
intrinsic drives these are things that
sort of you would generate and you
interact with a group based on those
things by contrast extrinsic motivation
is the external encouragement to do

something so this could be money paying
people this could be cost and benefits
so if there's a cost to
opening my data set I do it if there's a
benefit to opening my data set I will do
it and that leads to an inequality
within the scientific
community and then of course Prestige
which is of course if something is
prestigious I'll do it but if something
isn't prestigious I may not do it so it
interacts with intrinsic motivation but
it also is this external force that that
sort of is encouraging people to like a
you know a reinforcement mechanism
extrinsic motivations can also be where
things tie into existing
activities so one example is where a lot
of funding agencies and journals will
mandate that people share their data if
you want to publish a paper you have to
share your data so this is like tying
data sharing and doing this to
publishing the problem is there's no
intrinsic motivation to do this it's not
meaningful to everyone and so people
will do it to differing degrees of
completeness or of usefulness so if
you're forcing people to do something
versus it's something that people want
to do and think is a good idea it
becomes uneven and you don't have the
same quality
everywhere sometimes there's you know of
course there other types of mandates and
other T and there's been General uh an
overemphasis I think on mandates and
sort of top down motivation to do things

in open science and Open Access um
trying to get people to adhere to
certain standards or trying to enforce
more uh uh trying to enforce Moray that
may or may not fit the situation so this
is a problem in open science Open Access
and even open source
where where people you know you have to
kind of figure out the psychology of the
people doing
and enabling a lot of these open
practices so if we look at this graph we
see that their intrinsic motivations and
extrinsic motivations that interact so
at the middle of this in the gray box
you have intrinsic motivations things
like I want to do science I have Theory
building goals I need to create tools
and these things feed into extrinsic
motivations such as funding Community
Building uh adhering to the standards of
the field and adhering to rigor
standards of rigor so this is all kind
of these these intrinsic and extrinsic
motivations interact and then contribute
to how you carry out your open science
Open Access and eventually open
science so I'm going to shift to sort of
how we approach this in open
source and so this paper from neuron it
was published in
2017 it's called the commitment to open
science and neur open source it's called
a commitment to open source and
Neuroscience and uh P gon Andrew
Davidson R Angus silver and Georgio ESC
were the authors they're involved in an
organization called

incf which is the international neuro
informatics Corp and they've all kind of
done work in this area and they're
contributing to a project I'll talk
about in a little bit um that's open
source sort of soles of problems in
Neuro Imaging uh and how to do that so
one thing that they point out in this
paper is that code sharing is useful for
a number of things in
science but they make a claim that
leasing code can be used to replicate
results so one reason why we do code
sharing is so we can replicate our
results we can be confident that those
results can be replicated because we
have the code and everything is you know
perfect um would which is not always the
case but this is the claim this is the
motivator that they're talking about and
so sometimes you have to be strategic
about how you release your code
sometimes you know you just release your
code as a matter of you know requirement
but sometimes you can actually benefit
from releasing your code and in this CA
in the paper they talk about releasing
core scripts and libraries before
publication it will offer early feedback
from the community so you can release
your code just as a matter of sort of
requirement or you can benefit from it
by releasing your scripts and libraries
early so people can debug them and help
you if there are any problems with them
or whatever and so you end up with this
nice um optimal
situation another paper that I'd like to

highlight here is this paper from nature
technology a technology feature in the
journal Nature about six tips going
public with your lab software
so you know it's not enough to just
write the best quality programs and to
sort of put it out there you need to
really kind of think about a lot of
other things going on under the hood so
one of those things is that you need to
sort of have carve out sometime for
maintenance so if you put software out
into the world the reason you're one of
the reasons you're doing so is to sort
of serve your community you know
sometimes you want to be the standard in
the community other times you want to
give this GI this to other researchers
so that they do better
research sometimes you want The Prestige
of having like the top software in your
field but whatever the reason you want
to be able to maintain the software so
that it has a longetivity and so
different sources that they talk about
in this paper estimate anywhere from 5
to 15 hours a week to 20 30 20 to 30% of
the time of the project to be spent
maintenance maintaining the
software now as you might guess you know
there isn't a lot of motivation for that
unless you know your software is really
successful and so it's it can be a
problem if you you know take the leap
into putting your software out there but
don't devote the time to maintenance it
ends up hurting you
later another problem is that you need

to simplify runtime so a lot of times
you'll release software and people will
try to install it and fail because don't
have the right dependencies they don't
know how to run the software and
especially with a lot of academic
software we kind of build a clu of
plugins and other types of software on
top of software so that's not ideal and
that's a problem that can't be solved by
necessarily by in incentives and
motivations some of it is going to
depend on how we implement the software
and I'll talk about that in a little
bit we also made want to spend time in
interface design automated testing and
documentation to help users who you know
when you're in your own research group
you're sort of in a bubble where you
understand what you're doing you've used
the software you understand the specific
problem for which your software is
written well when you release it into
the public you might get people who are
not as experienced or simply don't
understand you know what some of the
variables are or whatever and so there
are a lot of ways that you know make
this easy to use is a is an
imperative and then a need for
transparency in community building so if
you really
software you need to have users to make
it worth your
while and if you have users you want to
build a community around that so that
you can do things like have a more you
know have a user base but also have a

community where you know you can solve
problems
collectively and address
open scientific problems and really for
everyone to benefit but also
transparency because we need to
understand everyone's methods and we
don't want to make it you know obtuse
for people when they use the software to
understand the methods that went into
generating the result that they're get
so you need to think about all these
things when you open up your software
when you go open source with academic
software
especially this brings us to another
point which which is the need for
research software Engineers so in
Academia we usually homebrew software we
end up putting it out there and you know
we maybe find time to maintain it we
find time to build a community but the
software is not up to the standards of
say professionally designed software and
so one of the things we don't have is
our research software engineering skills
in Academia a lot of academics know the
technicals matter but they don't know
how to build software so software needs
to be maintained and needs to undergo
quality control and so those things are
you know separate
skills so the problem of course is that
in Academia we tend to have graduate
students or open source contributors
build packages or whole software
platforms and their contributions are
well appreciated but their involvement

is itinerant so that means that they
might build something and then leave the
lab in two years where they may not you
know contribute to an open source
project and you may never see them again
and so they built something with no
documentation and obviously no
maintenance and it just kind of sits
there and Withers so I've seen this a
lot in my career where people will build
software for one purpose one use or like
for a single paper and then it goes away
because it's not maintained I mean near
the water of coat I've written when I
was a graduate student say where that
was the case and I wish that we could
recover that software and maintained it
but you know who has the time so really
what you need to do is you need to
include research software engineer roles
in your organization in your academic
organization and sometimes open- Source
projects will have this expertise but
sometimes they don't because there's no
requirement for it but it's also a
different skill set than say like the
academic skill set that people building
the software might have
so this is something where you can
either cross Trin people and there
programs where you can cross Trin
research software Engineers but more
often than not you just have to help
people acquire those
skills and so there are all sorts of
roles for RSC in different you know
academic and open- Source
organizations things like intentional

software design continuous integration
release Cycles bug fixes and maintenance
strategies and those are all things that
are separate skills from the academic
world separate skills even from like you
know if you even if you're computer
science researcher you're not
necessarily a software engineer so you
have to have that set of skills or at
least you know have a way for people to
acquire
them okay now we talk about open data so
open data is also a part of open science
and as as well as a part of Open Access
so Wikipedia defines open data as openly
accessible exploitable editable and
shared by anyone for any purpose so it's
a a data set that you can access you can
exploit or change you know you do
analyses on or whatever you can edit it
and you can share it and so anyone
should be able to do this now when they
say any purpose they mean that it's
licensed under an open license so an
open license is a permissive type of
ability protection and copyright for
people who put these things out there
that they work on and they want to
benefit from in some way but they also
want others to benefit from as well so
it's a it's a legal regime to maintain
you know to manage liability but also to
manage the social relationship between
the person making the data set producing
the data set and the person using
it open data allows for data to be
easily shared and reused but requires
govern issues of ownership and adherence

to community
standards so this you know part of this
is you know open data is in sort of
producing data sets for public
consumption the other part is working on
what they call an open data stack so an
open data stack is a management strategy
that is needed for sharing reusing And
archiving data so we typically start at
the bottom with some sort of interactive
interface for uh working with data
we need a metadata format we need a
Cloud Server to host our files on to
host our data and we also need standards
so we can have these higher level
standards for data
production for open uh science
production in general things need to be
findable accessible interoperable and
reusable so we need to have those four
principles embedded in our workflow and
then we need to have a strategy for
archiving data and for that we might use
something we dryad which is a a data
repository so we have these techn this
technology layer a standards layer
called security access layer and an
archival layer so put together that's
called a data stack and you want to
build a data stack that's you know with
tools that you know how to use and
standards that you know how to adhere to
but also things that will last you don't
want to pick something that will last a
year you have to migrate to something
else so you want a sustainable data
stack with sustainable tools with things
that are easy for your team to use and

so this also holds true for our Open
Access publishing strategy we don't want
to pick a strategy that isn't
sustainable because we'll stop doing
it now in the SE elegans community and
this involves my work with the open
roomm foundation you're fortunate in
that you have many open databases so the
neemod celegans is often used in
biomedical research and there's some
basic research on it as well and it's a
tight-knit community where there are
many many open data sets meaning people
produce data they share them uh with the
community and they have tools that are
curating a lot of these data sets and
papers and and things like that so we
have all sorts of data sets uh available
for all sorts of modes of inves a Sean's
biology and behavior so we have a number
of open databases covering Anatomy
genetics gene expression regulatory
elements Behavior
conomics so this includes like the worm
wearing site which includes data for
onod
conomics so we know in seans we know all
of the neurons in the conecto we know a
lot of their connections either through
Gap Junctions or through
synapses and we have worm wiring
diagrams for both the male and the
hermaphrodite so we have very rich data
on worm brains we also have worm Atlas
which is a atlas of structural Anatomy
so on top of the connectone data you can
fit structural Anatomy data and that's
at worm atlas.org the seans database uh

run by MDC in Berlin features a lot of data on proteomics and uh RNA okay and so this is another database that we have available senen is a database of gene expression and cgans in the nervous system so we can you know build a model of the connectome we can have the worm wiring data set senden data the worm Atlas the mdcc elegans database and we can actually you know produce a study from that so we have all those layers of data that we can include in and say a model or in an analysis we we also have this uh resource worm base which is where we can explore the world of seans and find mutants so seans has a number of defined mutants for specific genes they specific mutations that we can backcross and so if we have a strain of say some mutant defin mutant we have this worm that has a certain mutation that might be expressed in some Behavioral or anatomical deficit and so we can compare that to the wild type and say things about mutations of function and so this this is a really useful resource both for experimental work and for looking at the celan's genome and finding you know like alignments and other types of data like in one of papers we recently did we found uh different types of we aligned different uh open reading frames within gen so it's really a good resource and then of course uh nature methods this paper uh by people in the open worm Foundation um and this is called an open source platform for analyzing and

sharing worm Behavior data and this is
um on something called the turpy tracker
which is a movement tracker so they can
actually take microscopy images of
celegans they can track the worms
movement and they can characterize
different movements they they're
stereotypical movements worms make and
can actually track the behavior of the
worms and classify so that's you know
that that's the kind of data that's
available in celegans and so if you
think about the active influence
Community you know what kinds of data
sets and what kinds of resources could
be built around things
involving active inference so you might
for example think about you know and
some of these resources are just more
General resources that you could bring
to bear so you might say for example you
know go to a neuro Imaging resource and
then go to a behavioral resource or
build a behavioral resource there all
sorts of ways that you can build these
um open data repositories to help the
community Aid the community in solving
problems and answering
questions so this is a talk by ecof
freed this is called open science is an
antidote to poor practice cynicism so
one of the things he advocates for on
this video video is to use open science
as a way to combat poor practice like we
said open science is about reinforcing
practice and when you reinforce practice
of open science you reinforce good
scientific practice in general because

you're focusing on the process you're
focusing on you know cross checks and
and you know other things that involve
teamwork and so there there's a lot of
this sort of advocacy for helping
science through open science but also
combating cynicism where when we see
open when we see poor science we often
think well this is just the state of
science that it's actually failing or
it's bad and so you can fight
disinformation as well with open science
this is a nice set of views
here
so in the orthogonal lab we have done a
lot of work modeling open source
communities and their sustainability so
I talked earlier about Open Access
sustainab ility now I'm going to talk
about open source sustainability and so
we've done a lot of work with modeling
Frameworks to understand this
problem so why is sustainability
important well in the case of Open
Source many projects fail for a host of
reasons sometimes they're related to
people's projects you know not being of
good scope where they weren't asking
very good
questions sometimes the team falls apart
the team is dysfunctional
sometimes the project runs out of money
so there are a lot of reasons why
projects fail and sometimes open source
projects especially have problems with
maintainers problems with people
contributing so they fail for those
reasons as

well so you can have things like we're
scoping of a project we're trying to do
too many things we're trying to do too
few things we might have bad management
or a lack of
participation or we might have no
resources in terms of time money so
these are all things that we want to be
able to figure out how to manage at a
reasonable level and to do that we have
to understand
sustainability so in these you know in
these models that we've built we use
different
methods including active inference
reinforcement learning and large
language models to model open source
communities and understand what makes a
project
sustainable so the first one is this
pre-print on
that kind of includes a lot of the work
that we've done uh especially in terms
of reinforcement learning models and
active influence models this is our
research gate and then there's this new
work on com combining agent based models
and large language models this is Sarah
bosta walla's Google sumar code project
from this past year so she actually
presented in front of the Google sumar
code um uh Symposium and and she
presented her project there
and so this work so this work on large
language models involved using large
language models to generate issues and
then generate agents that could solve
those issues and see which kinds of you

know almost create fake projects and
then create issues to solve and then see
how the agents were able to solve them
and so you know we use the combination
of agent based techniques and large
language model techniques to sort of
work this a very exciting project
another approach is to take the
multi-agent reinforcement learning route
and so you see this grid down here and
this figure and all these dots and they
represent agents that are doing
different things sometimes they have
different statuses open source Community
sometimes they're of a different level
of Competency and you can actually run
these agent based models you can
Implement them in Misa as a python
package and run these agent based models
and then use reinforcement learning
algorithms to sort of optimize the
strategies that they're using so this is
you know again we are using these uh
sets you're creating these projects
you're using these sets of issues and
you're measuring their ability or their
competence to solve those problems and
so uh sh pomon RV Roger gopalan and hu
shogo were all involved in this work
over the past several years we've been
doing these kind of reinforcement
learning models for understanding open
source syst
sustainability then we've uh done this
work on Collective cognition for AI
ethics this is where I actually applied
active inference and we modeled agent
behavior in terms of active inference

using a package called interactively and
this is the GitHub repo here
interactively this applies a partially
observable partially observe markof
decision process or a pmdp model uh to
the problem
we also used a multi arm banded approach
to model explore and exploit behaviors
so explore and exploit is sort of this
tradeoff that agents make between
exploring a project exploring issues to
be solved and exploiting specific issues
solving them so it's like making you
know coordinating agent
behaviors and then figuring out what
strategy they're going to use and
modeling all of that so very interesting
project uh it needs more work on it so
if people in this group in in the active
inference Institute are interested and
following up on it that would be
welcome and so there are a number of
issues about maintenance that are very
important to sustainability and this is
a special in the academic sphere because
academic organizations and open source
typically don't have the resources that
private organizations might so
maintenance assist sustainability issues
are multitude in open source so I'll
talk about a couple of them here so the
first one is log4j and this happened a
couple years ago where there were a
number of Open Source projects that got
hit with the security threat and it was
because their security fell at a sync
with the current state-ofthe-art so in
open source we don't maintain things as

well as say commercial products so you
have these holes that can be exploited
by malicious actors so love 4J was so
serious that they had to um consult with
the US government to try to help fix
this problem and so this was a big
problem um with just kind of maintaining
the security of software but also it
exposed a big hole in the open source
model and so this is still something
that probably requires meta governance
you know some organization between like
the federal government and open source
organizations to
solve the second is called the make or
taker problem which is basically where
some organizations make open source
software and other organizations fork or
copy the software for their own purposes
thus being a taker and the problem here
is where that relationship is not
reciprocal that you know the makers will
develop systems that encourage
forks and then the open source
contributors or the takers are expected
to contribute back to the makers some
sort of you know maintenance or
something like that but they failed to
do so and so this is the problem where
you need to coordinate this reciprocal
relationship and so an example of this
is the word press for controversy so
WordPress is the maker in this case and
like I think it's WP engine is the taker
in this case WP engine was fored from
more press and it resulted in a fight
between the two
organizations one being the you know the

maker and one being the fork or The
Taker and they've been fighting over
commercial issues and fair use issues
that have served to break the Norms of
Open Source so there's this primary
issue of maintaining sort of a
reciprocal
relationship between organizations that
rely on basically rely on each other to
survive but you can have all sorts of
you know fights and disagreements about
different ways that the software can be
used commercially and other types of use
that you know are although are stated in
the license sometimes evolve away from
the license and their practical concerns
that you know isn't going to be solved
by just some sort of reciprocal
relationship or even intrinsic and in
extrinsic
motivations and then finally the final
example is XZ back door SL auto tools
this was an exploit that was made in in
an open source repository where it was
it resulted from a lack of maintainer
vigilance so this
maintainer in this particular case was
burnt out and they didn't feel like
really kind of closely inspecting every
poll request which is where a
contributor will submit a change or
contribution to an open source
organization and so ious malicious actor
was able to get exploits into the
software and it became a huge security
problem it's kind of like L forj and the
problem here is it's you know the
failure here is at the level of

maintainer and it wasn't the maintainer
fault necessarily it's that sometimes
people in open source organizations as
well as Open Access organizations open
sites have become burnt out especially
when you have to do all these things all
these practice oriented things to
realize open
vision and you know also you have
disincentives that lead to problem so if
you're not paying your maintainers
enough or if you're not Mo you know if
they don't have that intrinsic
motivation high enough they can't do the
prop the job that they need to do to
make things work or the extrinsic
motivations aren't well good enough you
don't really pay or maintain her very
much and therefore they're
disincentivized to sniff out exploits
and so these are our problems with
maintaining open source in an especially
in an academic
environment this is a second uh YouTube
video I wanted to point out this is
behind the scene behind the screen
there's a word play here which is open
source software is both a tool and topic
for research this is from William
Schuler at Santa Fe Institute and he
talks about sort of the history of
software and how you know you can
research open source software but also
use it as a tool it has a lot of
interesting issues in here so I
recommend watching this video now I'm
going to talk about how to sort of
implement open software for like a

purpose or an organization so we talked
about you know one of these points about
Distributing open academic software is
being being able to run it properly or
run it in a way that doesn't rely on too
many
dependencies and so this is where we can
use something like Linux
which or you know some flavor of Linux
to package data analysis tools so that
people can use them in an environment
where they don't have to worry about
dependencies they don't have to worry
about having the proper software
requirements so there are two examples
of where we can use a Linux dist to
package data analysis tools and present
them to people in the community
sometimes with fairly low technical
standards
so the first is narof Fedora and so this
was created as free and open source
software this is where we have a version
of the Fedora this Dr of Linux and we in
Fedora you package a bunch of
neuroimaging analysis tools and you put
them in and you just give people the
package they install it on their machine
and then they can run those tools
without worrying about dependencies or
getting thing onun on Windows or Mac or
whatever
the other example is neurod Debian this
is run by incf this is the group led by
por gleon and and Angus or
Angus um run by run by or Leon and and
colleagues and this is where they have a
duby and drro of Linux and they have

these neuroimaging tools in Debbie and so you have Fedora and Debian you can choose between the two dros and you can run these enging Tools in these environments and that's that makes it easier I think for neuroscientists to do this kind of work to work with these packages uh similar thing could be done for active inference and especially in Fedora you have these different flavors of Fedora like for artists or for musicians where you can package tools for those professions and they can just you know use it out of the box so focusing on the neurop Fedora uh this is a special interest group whose goal is to provide Neuroscience researchers and enthusiasts with a strong easy to use readymade free and open source software platform for their work and anchor Sinha has let the taken the lead under aladora Morgan Huff is part of the orthogonal lab has also worked on this as well then we have neurodes so neurodes is not an operating system but it's a virtual machine it's a virtualization or a containerized environment where you have like basically virtual machine on your existing computer and you run everything inside of that and it really paves over a lot of the operating system differences and system requirement differences that you might find trying to install open source software on your home computer so like you know this is where you've basically created a controlled environment for the software

so it really helps people run it when
they have very little Tech technical
expertise and just get it to run that's
all you need to do so neurodes uh is a
containerized data analysis environment
specialized for neuroimaging research
and this is something that you could
easily do with the active
inference it's where you package the
software code with the operating system
libraries and dependencies required to
run the code in a single lightweight
executable so this is platform agnostic
it's just you open up the container you
run it and you can do everything inside
of
there so thank you for your attention um
I especially want to thank the members
of the open source interest group where
Bradley Alisa Jesse parent Morgan Huff
Sarah Bala Brian mccorkel amanu shle rvy
Roger gopalan and Shu
pamon um we have other people who cycle
through the group from time to time and
you're welcome to join
us uh please join the worth going a
research lab and then be on the lookout
for open- Source activities we usually
do the interest group in the summer if
you're interested get in touch otherwise
Happy active inferring